Anantha Lakshmi
INSTITUTE OF TECHNOLOGY & SCIENCES
(AUTONOMOUS)

ACADEMIC PROJECT · 2026–27

StudyMapper

AI-driven personalised learning and academic management

TEAM

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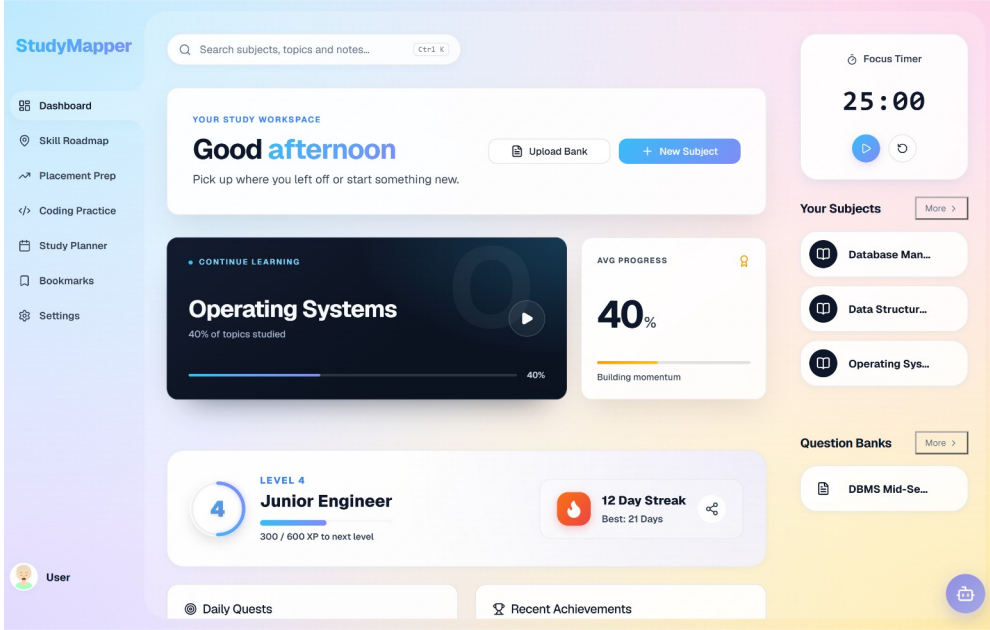
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GUIDE

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The running build — dashboard at 1440 × 900

Live demo · study-mapper.vercel.app

Every number and screen in this deck comes from the repository or the running app.

Ten slides, one story

Why it exists → what it does → how it works → what is proven → what comes next.

01 **Title & team**

02 **Agenda**

03 **Abstract & motivation**

04 **Problem & objectives**

05 **Literature survey**

06 **System architecture**

07 **Methodology & algorithm**

08 **Implementation details**

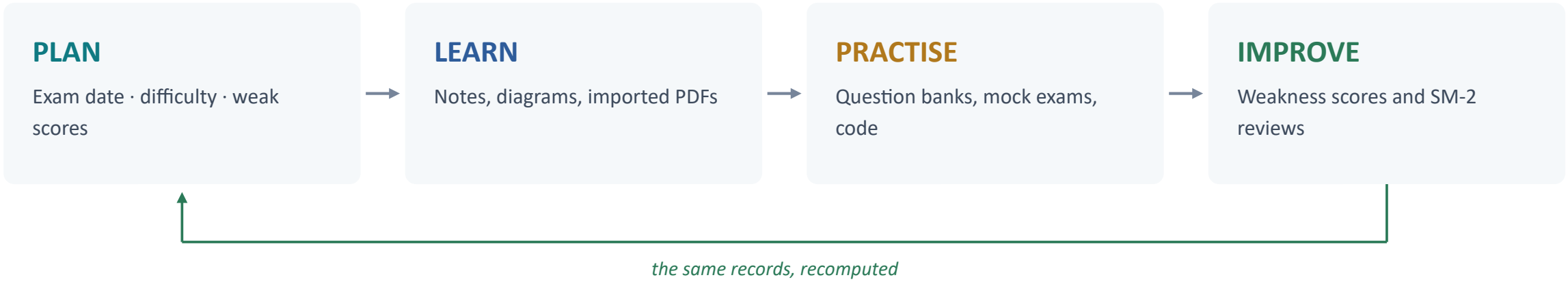
09 **Results & analysis**

10 **Conclusion, future & references**

Short version: about ten minutes, then questions.

One loop: plan, learn, practise, improve

A syllabus becomes a plan that reacts to what the student actually gets wrong.



WHY IT MATTERS

Fixed study time
The same hours for every topic, whatever its difficulty.

Late feedback
Mistakes surface at the exam, not during practice.

Scattered tools
Notes, deadlines and tests live in different apps.

The gap is response, not content

Students do not lack material. They lack a system that reacts to their week.

WHAT IS BROKEN

01

Fragmented tools

Notes, deadlines and tests live in different apps.

02

Static plans

One plan for every topic, whatever its difficulty.

03

Late feedback

Mistakes surface at the exam, not during practice.

WHAT WE SET OUT TO DO

✓ Adaptive plan

Recomputed from the exam date, difficulty and weak scores.

✓ Generated material

Notes, diagrams and questions the student can edit.

✓ Measured weakness

A 0–100 score per topic, from the last seven days.

Out of scope · institutional timetabling · proctored exams · a native mobile app

Five approaches, one missing loop

Each category solves part of the problem. None closes the loop for one student.

	Personalises	Uses own data	Works offline	Closes loop
Adaptive platforms	●	●	—	○
AI tutors	●	○	—	—
LMS analytics	—	●	—	—
Spaced repetition	●	●	●	—
Gamified apps	○	○	●	—

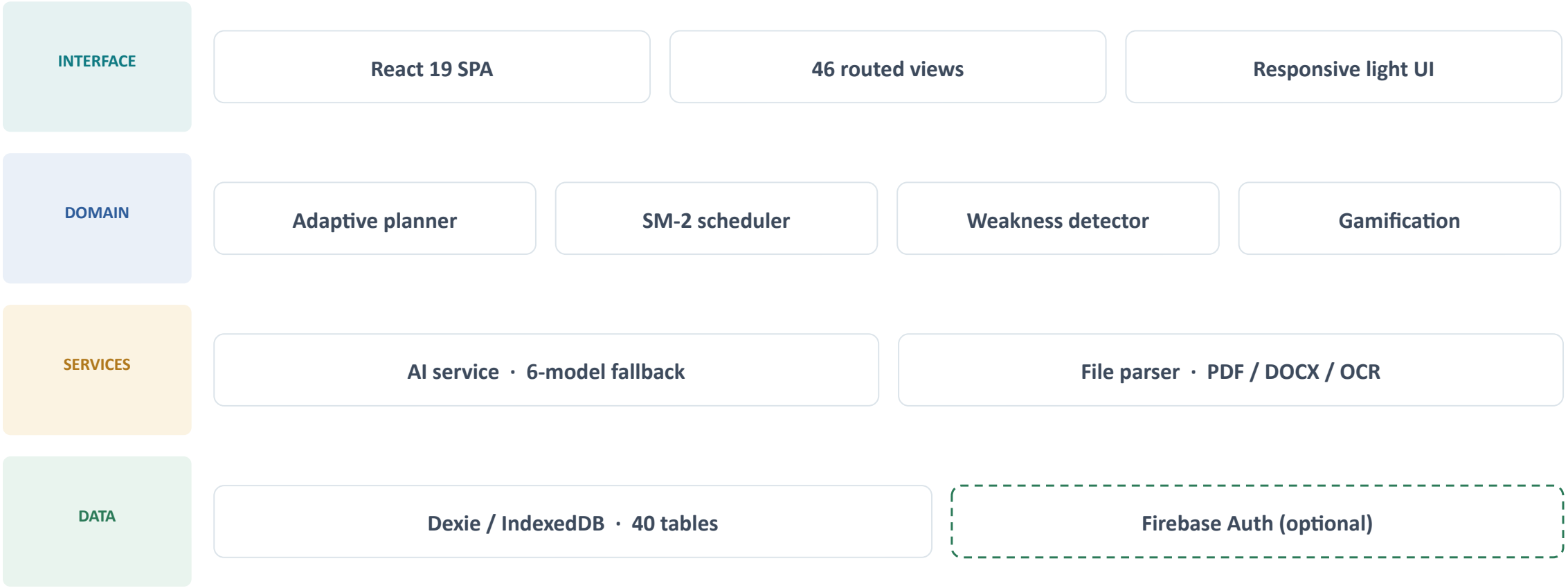
THE GAP **No category combines planning, practice, measurement and offline study for one learner.**

● built in ○ partial — absent

Design comparison, not a systematic review.

Four layers, two optional services

Everything that matters runs in the browser. The network is optional.



Solid boxes run locally · the dashed box needs the network · no server in the study loop.

From activity to next action

Every study action is scored, and the score changes the plan.

01 **Log activity**
Attempts, pomodoros, notes and coding runs.

02 **Score each topic**
A 0–100 weakness score, computed locally.

03 **Flag the weak**
40 or above, over the last seven days.

04 **Re-plan**
Regenerate the plan from the latest scores.

Fixed weights, inspectable rules — no training, no black box.

WEAKNESS SCORE

0–100 per topic, clamped

+30	failed coding attempt
+0.5 × (100 − accuracy)	low accuracy, below 60
+10	completed pomodoro
−20	successful coding attempt
−10	notes generated

Flagged as a weakness at 40 or above.

SM-2 **interval ← interval × ease factor**

A browser app, not a server stack

Four groups of tools, chosen to keep the loop small and free to run.

UI

Interface

React 19

Vite

React Router

Fast feedback for a small team.

AI

Intelligence

OpenRouter

Pyodide

Mermaid · KaTeX

No model training or hosting.

DB

Data

Dexie

40 tables

Firebase Auth

Local data; sign-in is optional.

✓

Quality

Vitest · 47 tests

Vite build

Vercel deploy

Every claim is checked.

HARDWARE & RUNTIME

Any modern browser. No GPU, no server, no model hosting, no paid tier.

The same build runs on a laptop and on the deployed demo.

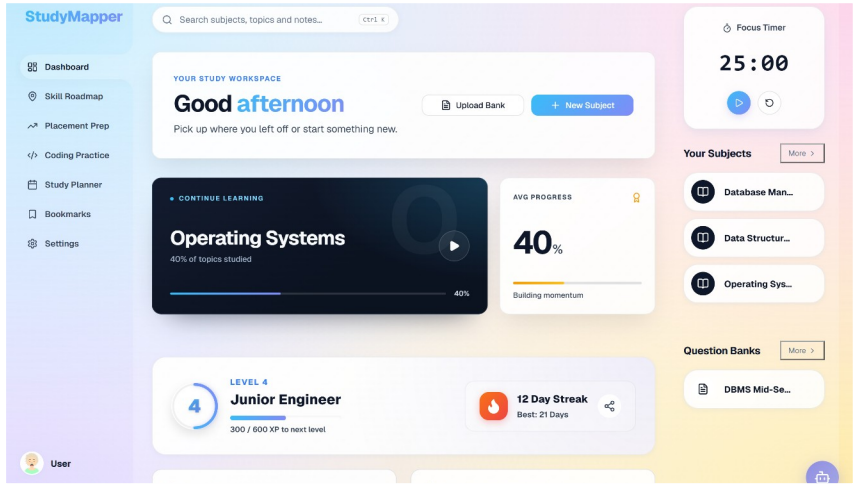
What is verified — and what is not

Functional completeness and build health, measured. Learning gains are the next study.

47/47	6.21 s	40	46
automated tests	production build	database tables	routed views

BEFORE → WITH STUDYMAPPER

Four apps	→	**One workspace**
Plan made once	→	**Plan recomputed**
Feedback at the exam	→	**Feedback in practice**



Dashboard — the seeded demo build

Not claimed: a measured learning-outcome study, institutional deployment, a teacher dashboard, a native app.

Three claims, one next step

The loop is built and verified. Whether it improves learning is the experiment ahead.

ADAPTIVE

Plans recompute from the exam date, topic difficulty and measured weakness.

Regenerate on demand; panic mode compresses the rest.

EXPLAINABLE

Every score traces back to stored attempts; the rule set is documented.

Flagged at 40+, over a seven-day window.

LOCAL-FIRST

Notes, plans and attempts live in IndexedDB; AI and sign-in are optional.

No server is required for the core loop.

NEXT

01 **Classroom pilot**
Pre/post quiz with a real cohort.

02 **Optional sync**
Account-based, conflict-safe sync.

03 **Accessibility pass**
Keyboard paths, contrast, screen readers.

REFERENCES

[1] StudyMapper source repository, GitHub.

[2] StudyMapper project abstract, ALITS, 2026–27.

[3] React 19, Vite and Dexie documentation.

Full reference list with access dates in the detailed deck.

[4] OpenRouter API documentation.

[5] P. A. Woźniak, SuperMemo 2, 1987.

[6] Vitest and Pyodide documentation.